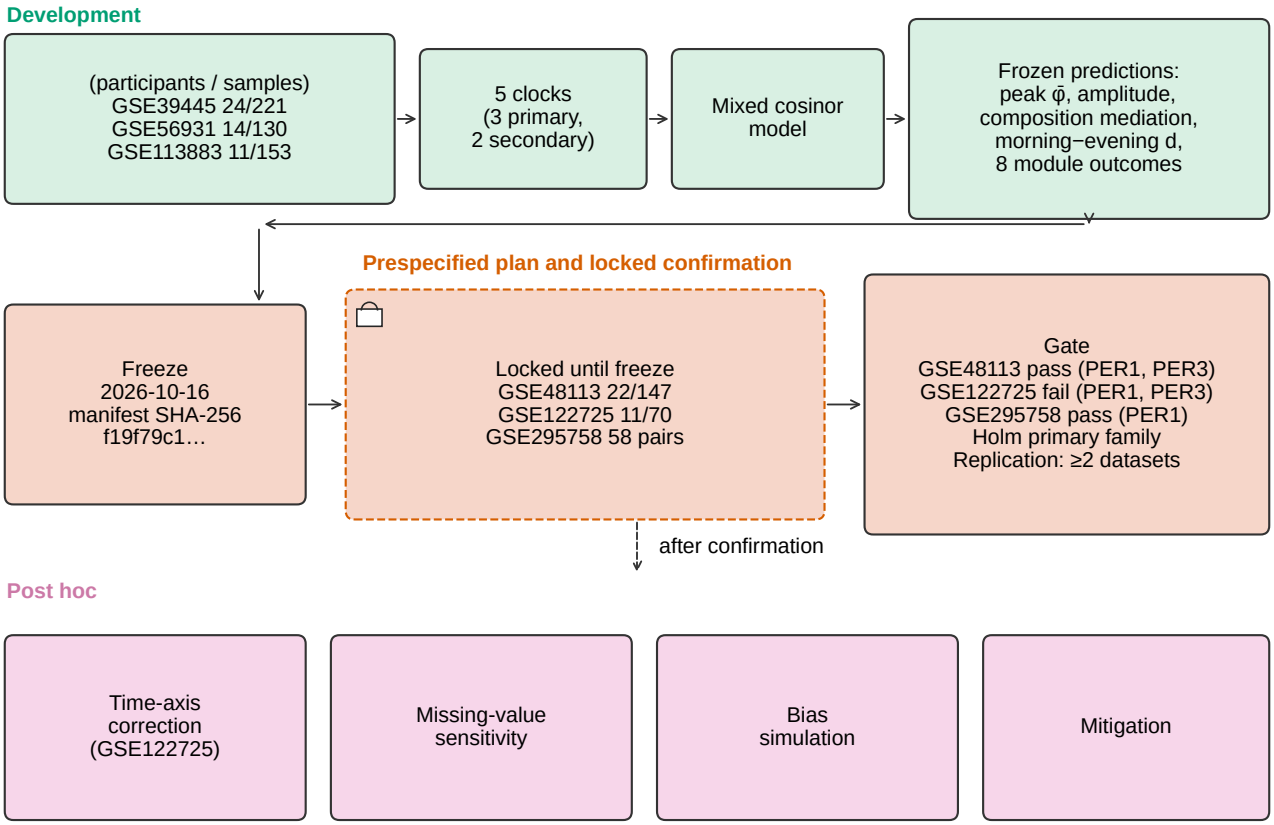
Development
(participants / samples)
GSE39445 24/221
GSE56931 14/130
GSE113883 11/153
5 clocks
(3 primary,
2 secondary)
Mixed cosinor
model
Frozen predictions:
peak φ̄, amplitude,
composition mediation,
morning−evening d,
8 module outcomes
Prespecified plan and locked confirmation
Freeze
2026-10-16
manifest SHA-256
f19f79c1…
Locked until freeze
GSE48113 22/147
GSE122725 11/70
GSE295758 58 pairs
Gate
GSE48113 pass (PER1, PER3)
GSE122725 fail (PER1, PER3)
GSE295758 pass (PER1)
Holm primary family
Replication: ≥2 datasets
after confirmation
Post hoc
Time-axis
correction
(GSE122725)
Missing-value
sensitivity
Bias
simulation
Mitigation